# ITLDO801 Assignment 3

**Student:** Kevin Abijuru
**Repository:** kevinabijuru.github.com
**Module:** Development Operations (ITLDO801)
**Topic:** Test Logging and Reporting with Selenium

## 1. Test Plan

| Test ID | What is tested | Steps | Expected result |
|---|---|---|---|
| TC-01 | Home navigation link | Open the portfolio, locate Home, click it, inspect URL. | URL path is /index.html. PASS. |
| TC-02 | About navigation link | Open the portfolio, locate About, click it, inspect URL. | URL path is /about.html. PASS. |
| TC-03 | Services navigation link | Open the portfolio, locate Services, click it, inspect URL. | URL path is /services.html. PASS. |
| TC-04 | Portfolio navigation link | Open the portfolio, locate Portfolio, click it, inspect URL. | URL path is /portfolio.html. PASS. |
| TC-05 | Contact navigation link | Open the portfolio, locate Contact, click it, inspect URL. | URL path is /contact.html. PASS. |
| TC-06 | Portfolio Details in Pages menu | Open the portfolio, open Pages, click Portfolio Details, inspect URL. | URL path is /portfolio-details.html. PASS. |
| TC-07 | Missing-link negative test | Search for the deliberately absent missing-assignment-link.html link. | Locator failure is handled and recorded as FAIL with URL, error, and screenshot. |

## 2. Implementation

> The Selenium runner uses Python's logging library with a unique timestamped log per run. It records run headers, browser version, operating system, test IDs, expected and actual results, status, duration, failure URLs, errors, and screenshots.

- **Script:** selenium_tests.py
- **Dependency:** selenium
- **Container:** Dockerfile and docker-compose.yml with headless Chromium
- **Evidence folder:** logs/

## 3. Verification Evidence

Three fresh container runs were completed. Each run recorded six passing positive tests and one controlled negative failure.

| Run log | Result | Failure evidence |
|---|---|---|
| selenium_run_20261025_145153_692154.log | 6 PASS, 1 FAIL | TC-07 screenshot |
| selenium_run_20261025_145231_379973.log | 6 PASS, 1 FAIL | TC-07 screenshot |
| selenium_run_20261025_145256_533847.log | 6 PASS, 1 FAIL | TC-07 screenshot |

Generated screenshots are stored under `logs/screenshots/`. The negative test is intentional and demonstrates that the suite continues after a failure.

# 4. Reproduction

```
docker compose build
docker compose run --rm selenium-assignment
```

The generated logs are not manually typed. They are produced by the Selenium script during each execution.